Human-Computer Interaction

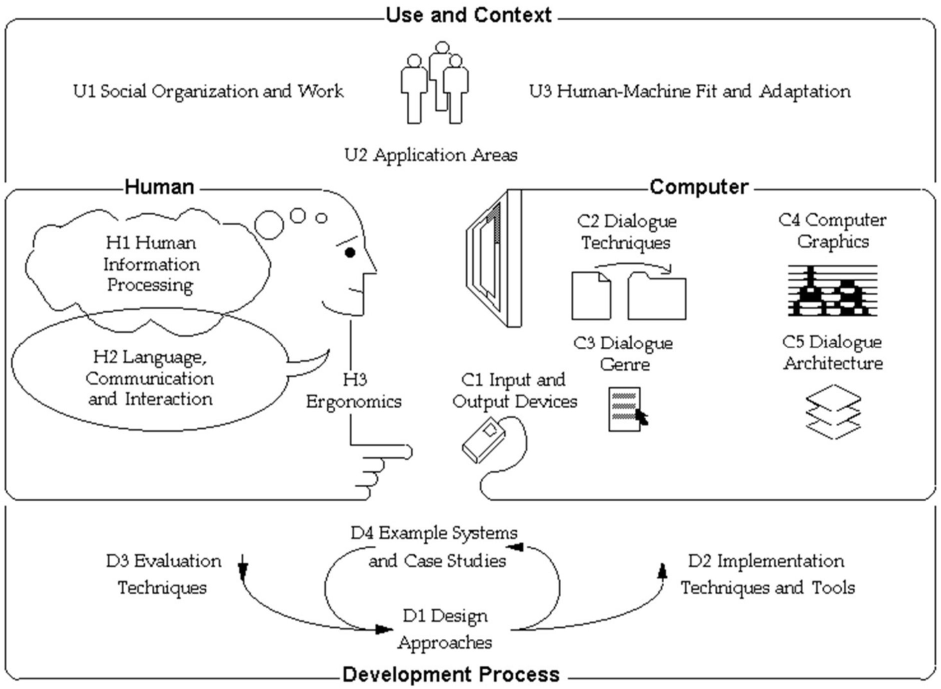
Johan F. HOORN

Lecture 01: Use and Context (continued)

Course overview

Lecture topics:

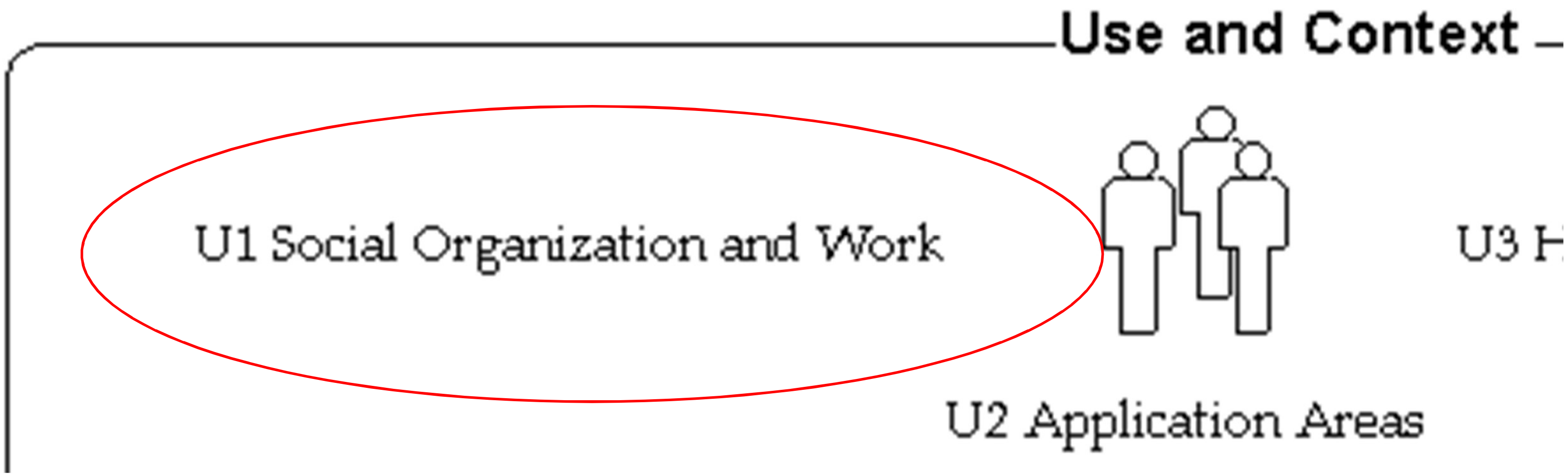
1. Use and context	U1-3
2. Human information processing	H1
3. Visual perception and computer graphics	H1, C4
4. Language, communication, and dialogue techniques	H2, C2,3,5
5. Ergonomics, I/O devices, haptics, sound	H3, C1
~~Design approaches and implementation~~	~~D1-2~~
6. Evaluation techniques and example systems	D3-4



http://www.acm.org/sigchi/cdg/

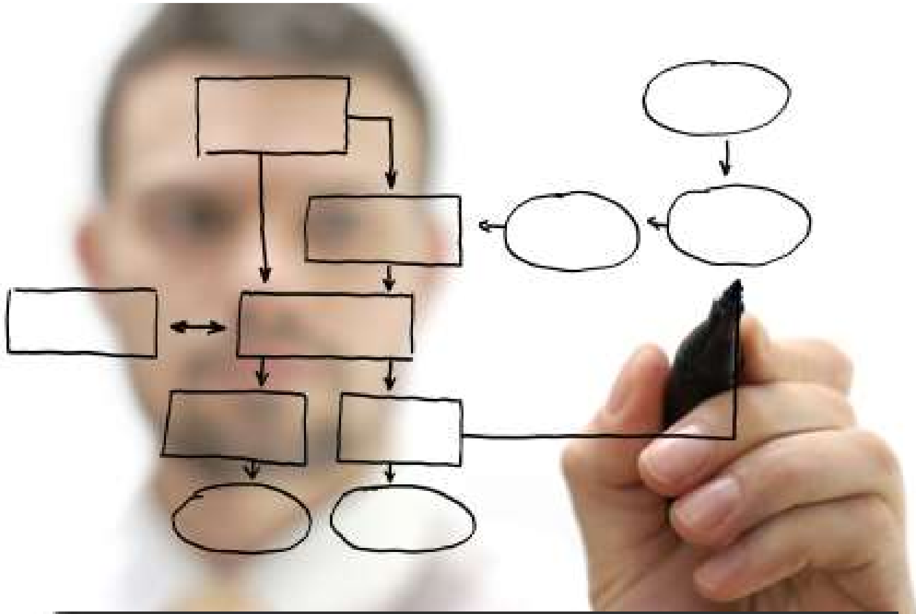
Model of HCI
SIGCHI (1992)

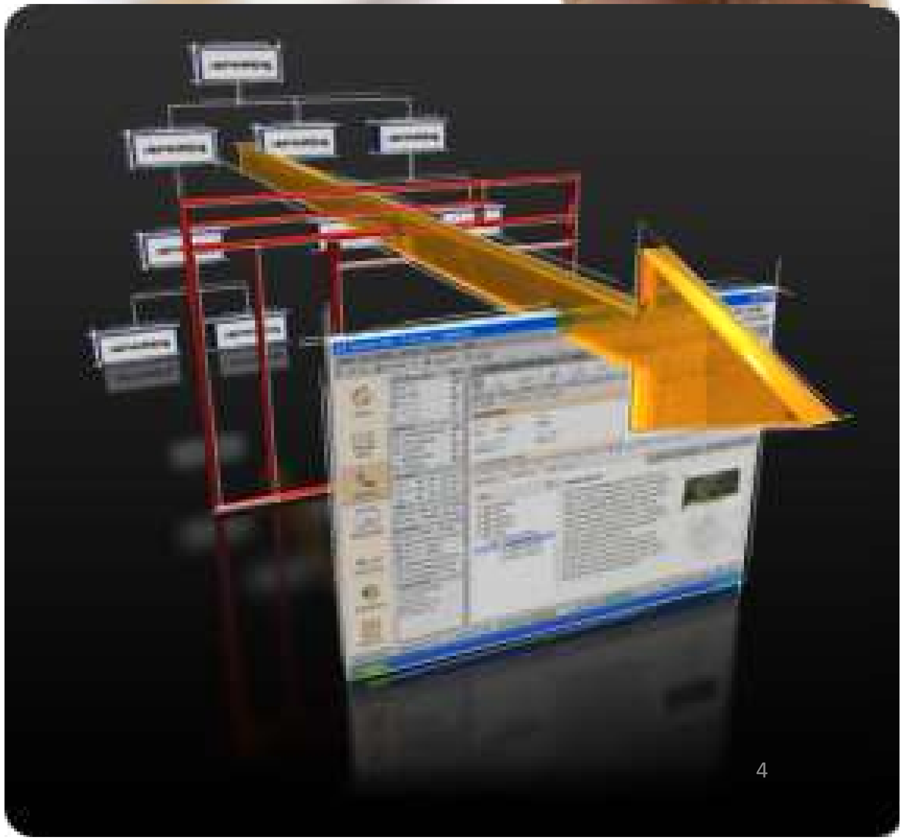
HCI Today
is about
Use and Context



Multidisciplinary

- Computer science
- Artificial intelligence
- Engineering
- Cognitive psychology
- Ergonomics, human factors
- Sociology
- Communication
- Anthropology
- Linguistics
- Industrial design
- &c





Former students sometimes write me

Back-end audit administrator at Sina.com

“You go through every video and message people post. The most depressing part is the algorithm in Sina.com cant well discern the meaning of those messages. So in the last, some of the tweets need to be checked manually.”

Creative Coders in Organizations

Or how to make these people work together?

Contents

☯ Different work preferences and values
☯ Different use of their brains
☯ Different problem types they solve with it

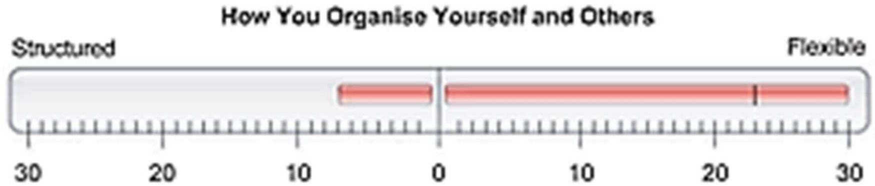
☯ The project life cycle (the Innovation S-curve)
☯ Plotting work preferences and brain structures onto the S-curve
☯ Distributed leadership among team members

☯ How companies are organized (even if they say they aren't)
☯ Plotting the S-curve onto organizational structure
☯ The role of communication networks

☯ Analyze a number of real-life cases with me

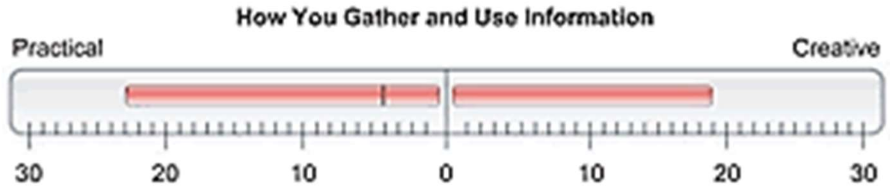
☯ Different work preferences

Margerison-McCann
Team Management Systems (TMS)



This person is more likely to thrive in an environment without a rigid structure and requiring a more flexible approach

This person is likely to enjoy taking ideas through to reality



☯ Different work preferences

This person has a strong preference for organizing, developing and producing and is less likely to move outside their comfort zone

This person is likely to enjoy working in many different areas and show more versatility

☯ Different work values

Margerison-McCann
Team Management Systems (TMS)

Work values

1. Individualism: Focusing entirely on personal drivers

2. Collectivism: Focusing more on the group than the individual

3. Compliance: Needing to work with an agreed set of rules and procedures

4. Empowerment: Requiring autonomy and needing personal input

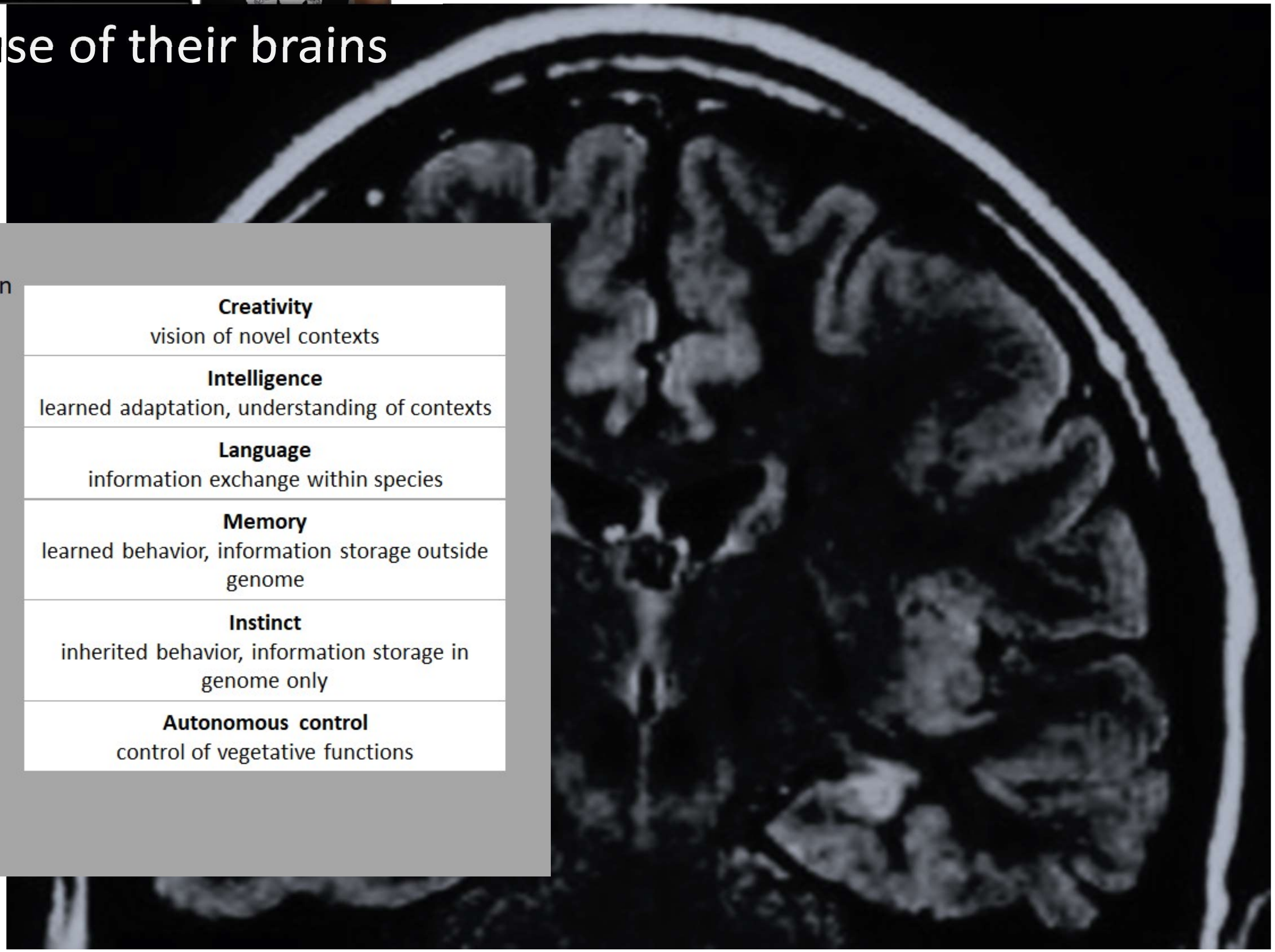
5. Authority: Wanting to work with clear systems

6. Independence: Valuing personal creativity and independence

7. Conformity: Avoiding extreme action and conforming to external expectations

8. Equality: Focusing on both the group and organisational freedom

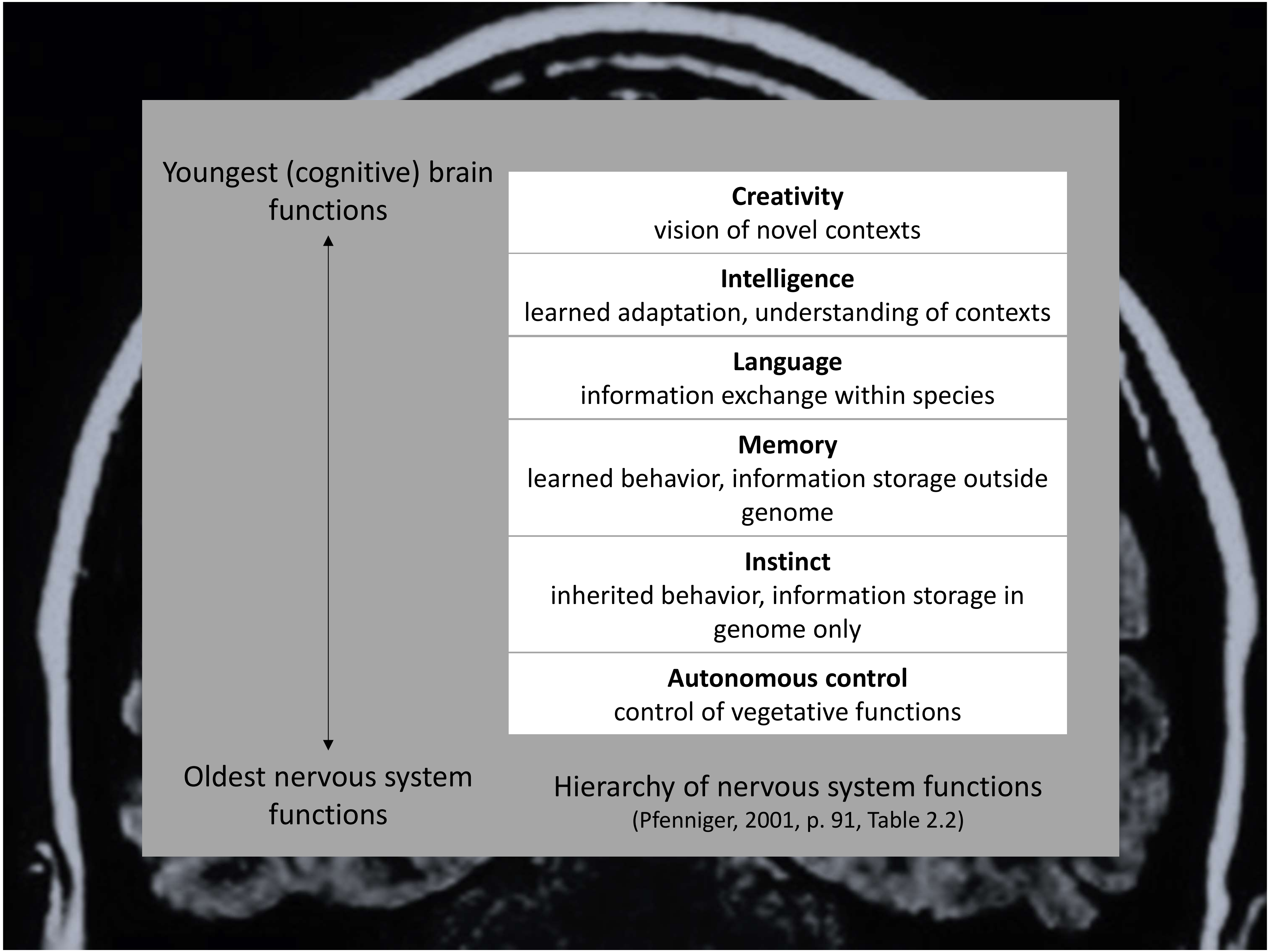
☯ Different use of their brains



Youngest (cognitive) brain functions

Creativity vision of novel contexts
Intelligence learned adaptation, understanding of contexts
Language information exchange within species
Memory learned behavior, information storage outside genome
Instinct inherited behavior, information storage in genome only
Autonomous control control of vegetative functions

Oldest nervous system functions



Youngest (cognitive) brain functions

Creativity vision of novel contexts
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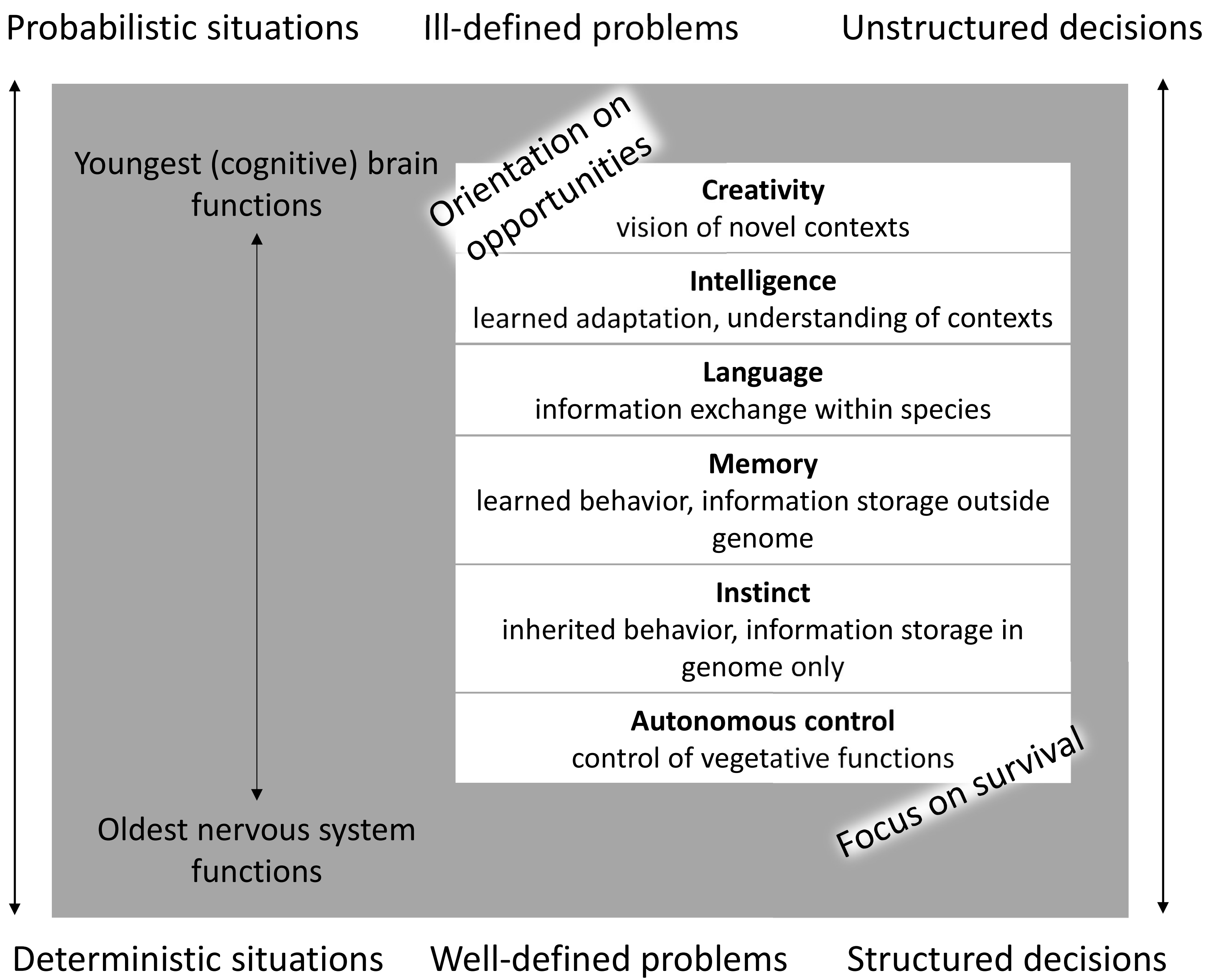
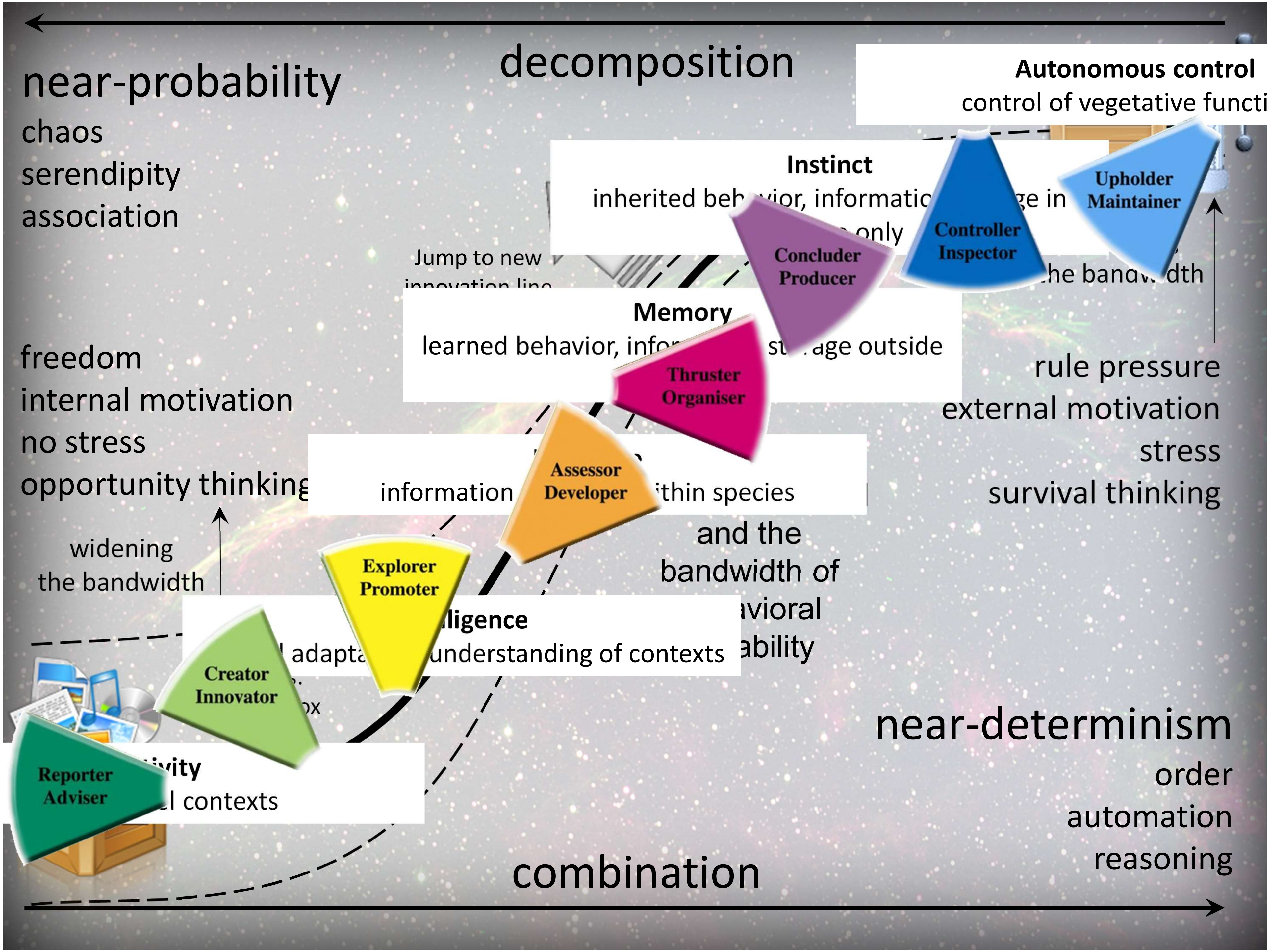
Oldest nervous system functions

Hierarchy of nervous system functions
(Pfenniger, 2001, p. 91, Table 2.2)


Probabilistic situations
Ill-defined problems
Unstructured decisions
Youngest (cognitive) brain functions
Orientation on opportunities
Creativity
vision of novel contexts
Intelligence
learned adaptation, understanding of contexts
Language
information exchange within species
Memory
learned behavior, information storage outside genome
Instinct
inherited behavior, information storage in genome only
Autonomous control
control of vegetative functions
Focus on survival
Oldest nervous system functions
Deterministic situations
Well-defined problems
Structured decisions


decomposition
near-probability
chaos
serendipity
association
Autonomous control
control of vegetative functi
Instinct
inherited beh vior, informatic ge in
only
Jump to new
Concluder
Producer
Controller
Inspector
Upholder
Maintainer
he band dth
Memory
learned behavior, info st age outside
Thruster
Organiser
freedom
internal motivation
no stress
opportunity thinking
rule pressure
external motivation
stress
survival thinking
information
Assessor
Developer
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Explorer
Promoter
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Creator
Innovator
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Reporter
Adviser
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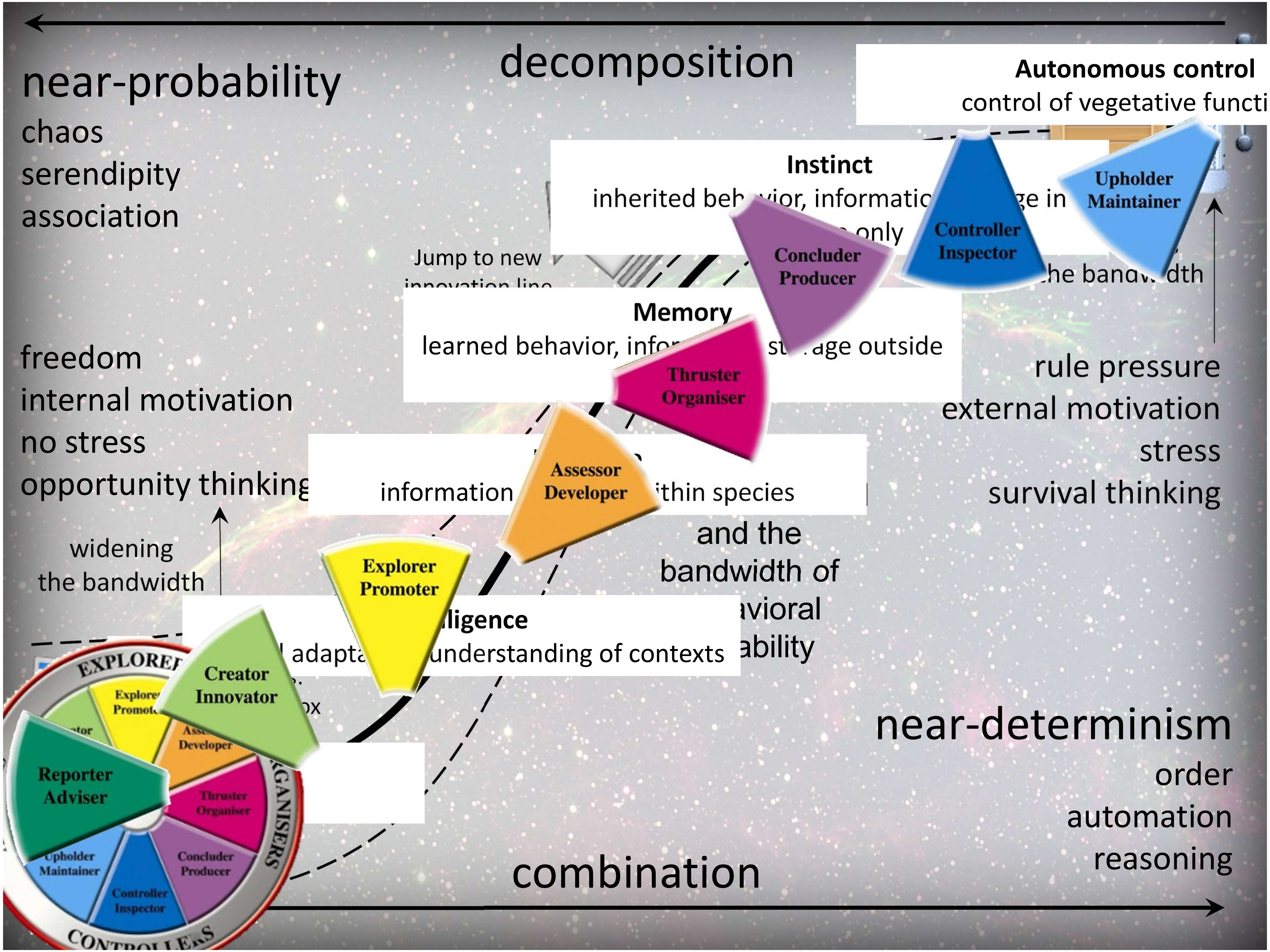
near-determinism
order
automation
reasoning
combination



decomposition
near-probability
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Autonomous control
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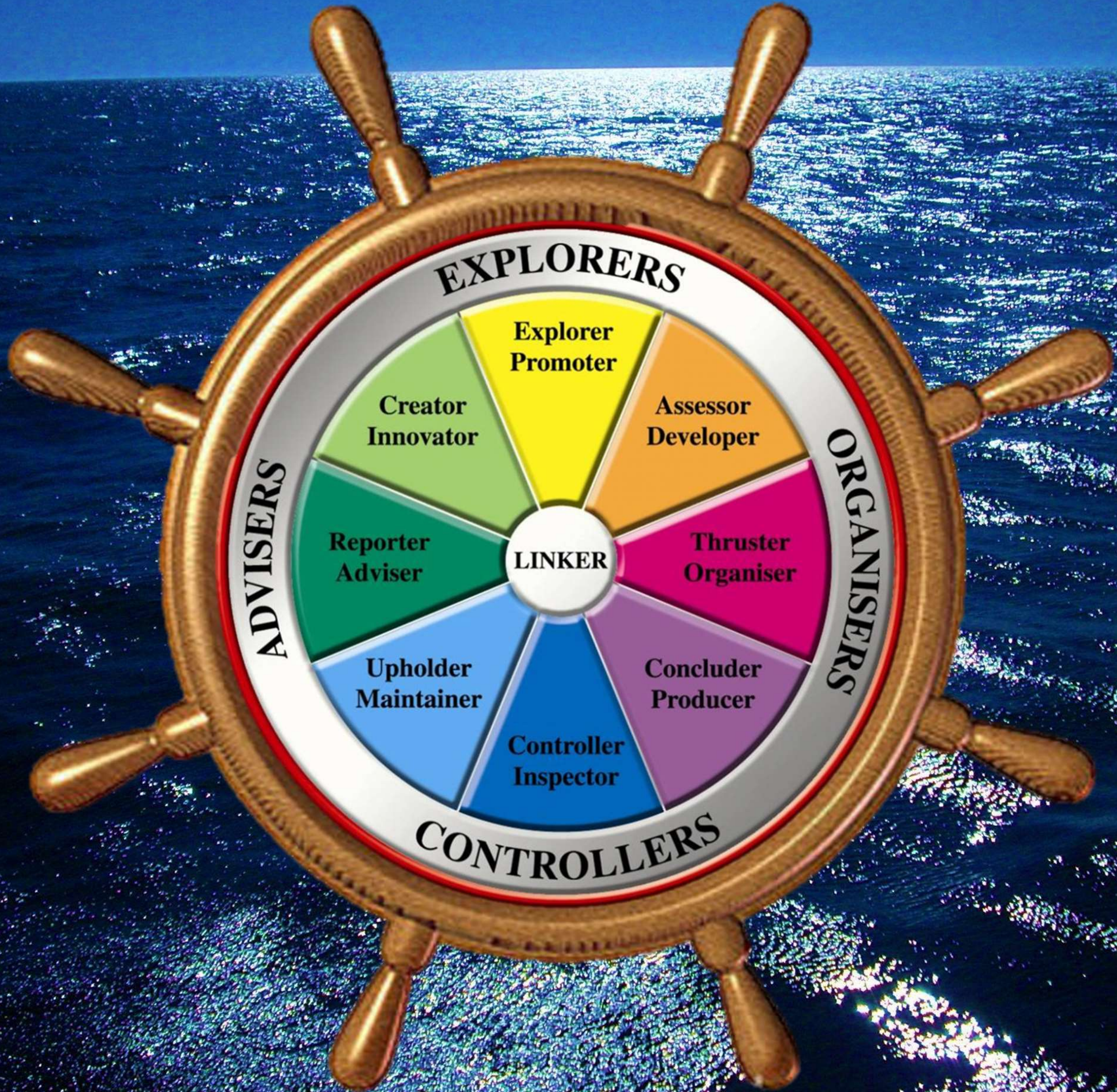
combination
Upholder Maintainer
Controller Inspector
Concluder Producer
Thruster Organiser
Assessor Developer
Explorer Promoter
Creator Innovator
Reporter Adviser
EXPLORER
ORGANISERS
CONTROLLERS

Distributed leadership among team members



☯ Different problem types they solve with it

- From ill-defined probabilistic to well-defined deterministic

☯ The project life cycle (the Innovation S-curve)

- The type of problem evolves from probabilistic to deterministic

☯ Plotting work preferences and brain structures onto the S-curve

- At each stage another type of problem solver is required and…

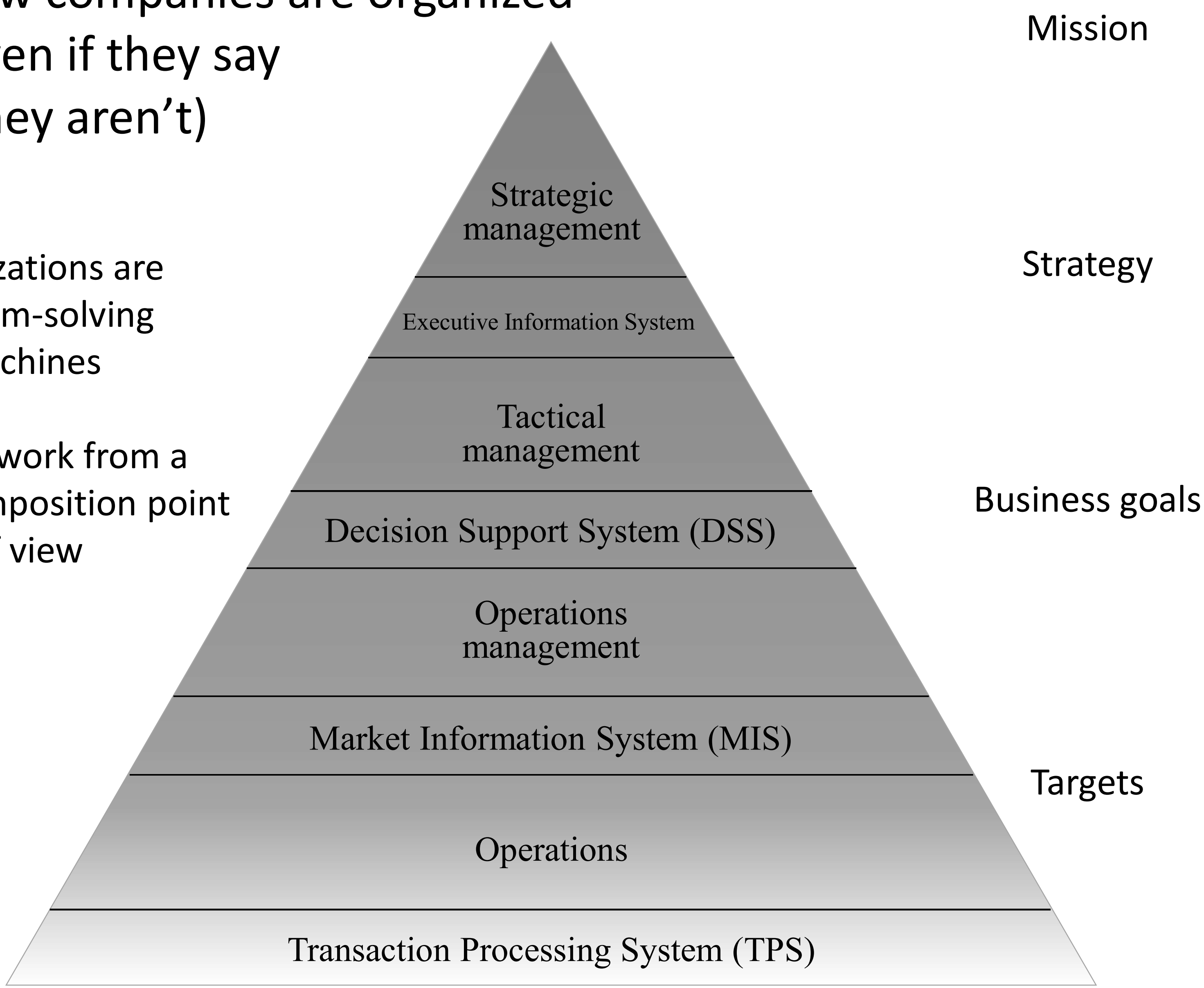
☯ Distributed leadership among team members

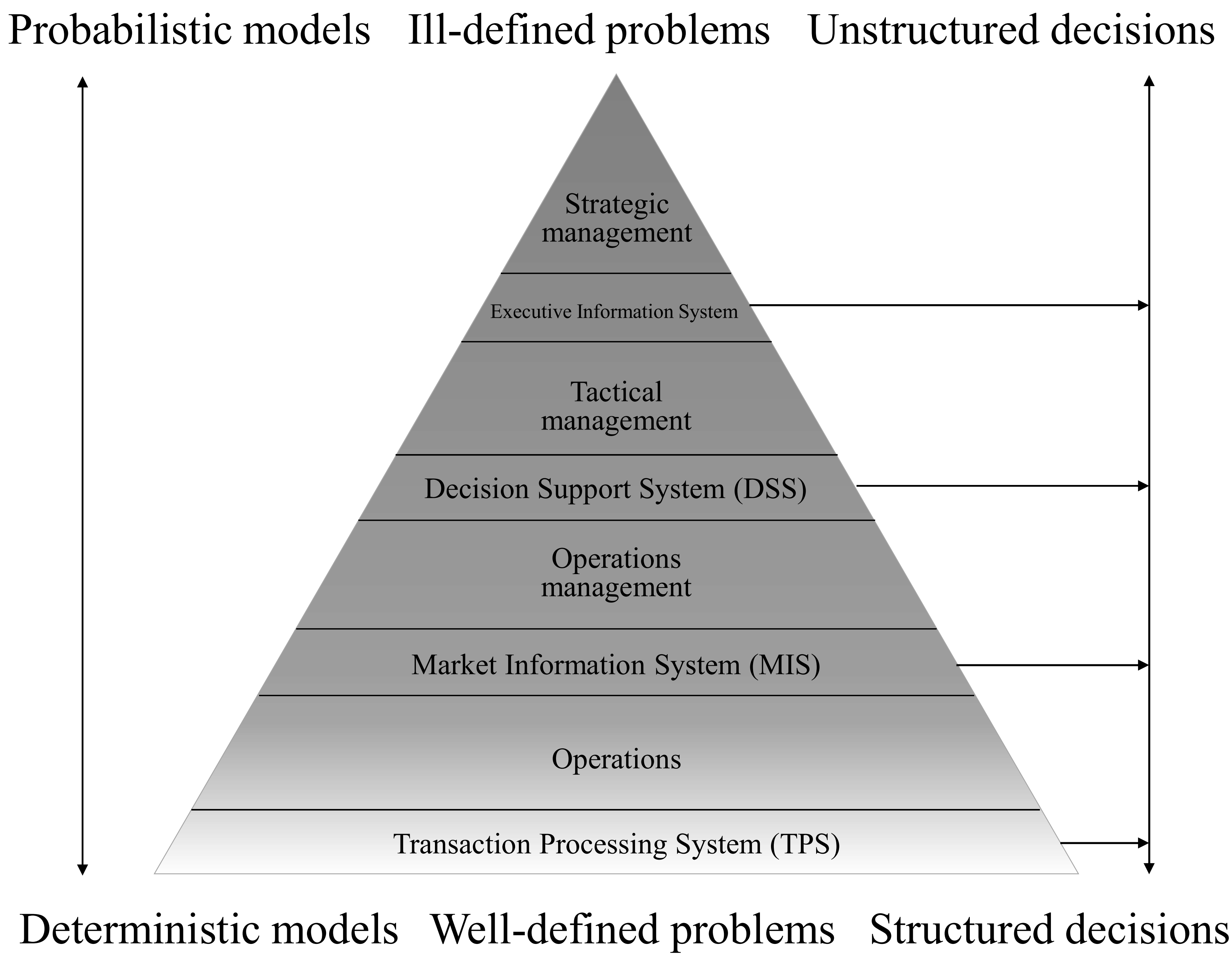
- … should take the lead (no one is ‘the boss’ and no one should be shy)

☯ How companies are organized (even if they say they aren't)

Organizations are problem-solving machines

They all work from a goal-decomposition point of view

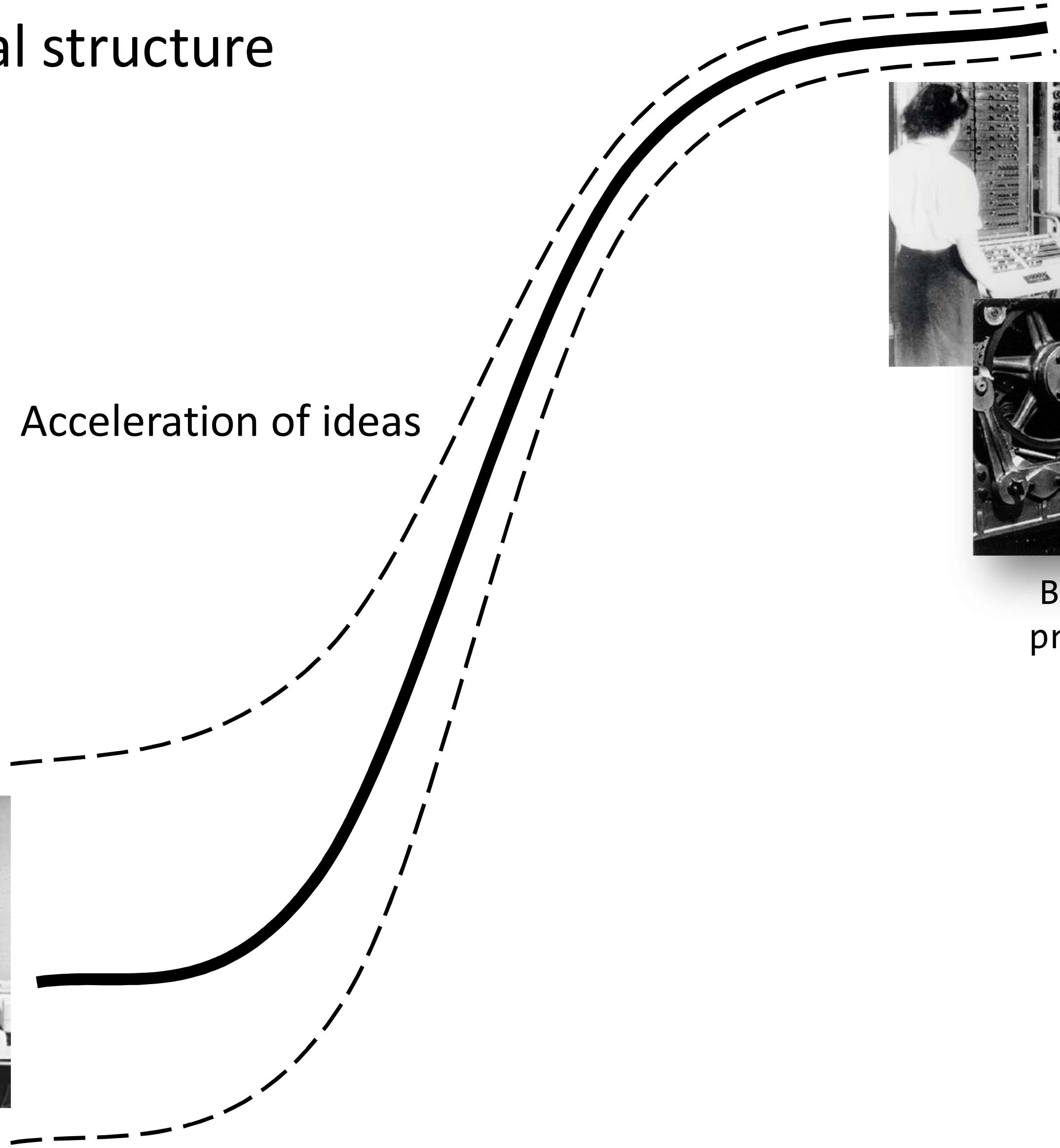


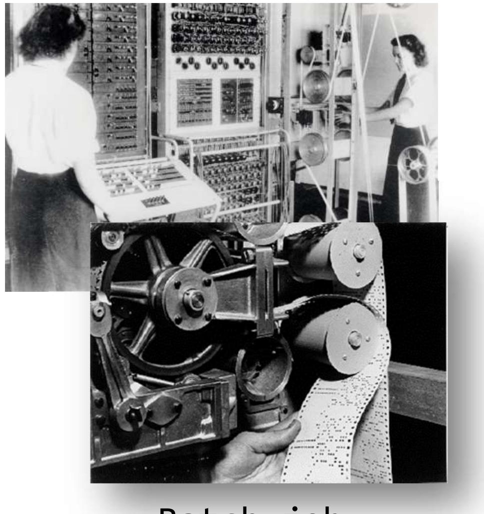


Probabilistic models
Ill-defined problems
Unstructured decisions
Strategic management
Executive Information System
Tactical management
Decision Support System (DSS)
Operations management
Market Information System (MIS)
Operations
Transaction Processing System (TPS)
Deterministic models
Well-defined problems
Structured decisions

☯ Plotting the S-curve onto organizational structure

Maturity of ideas

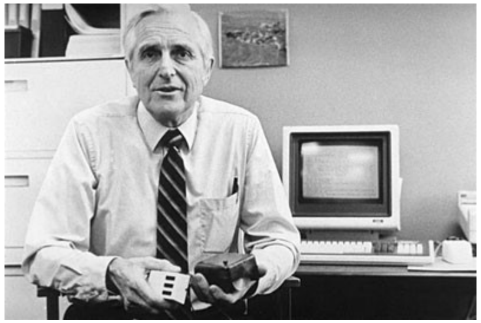




Batch job processing

Acceleration of ideas

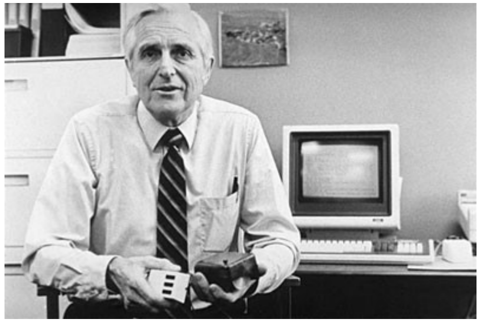
Douglas Engelbart



Infancy of ideas

A Conceptual Framework for
Augmenting Human Intellect

Crazy ideas



The new
vision

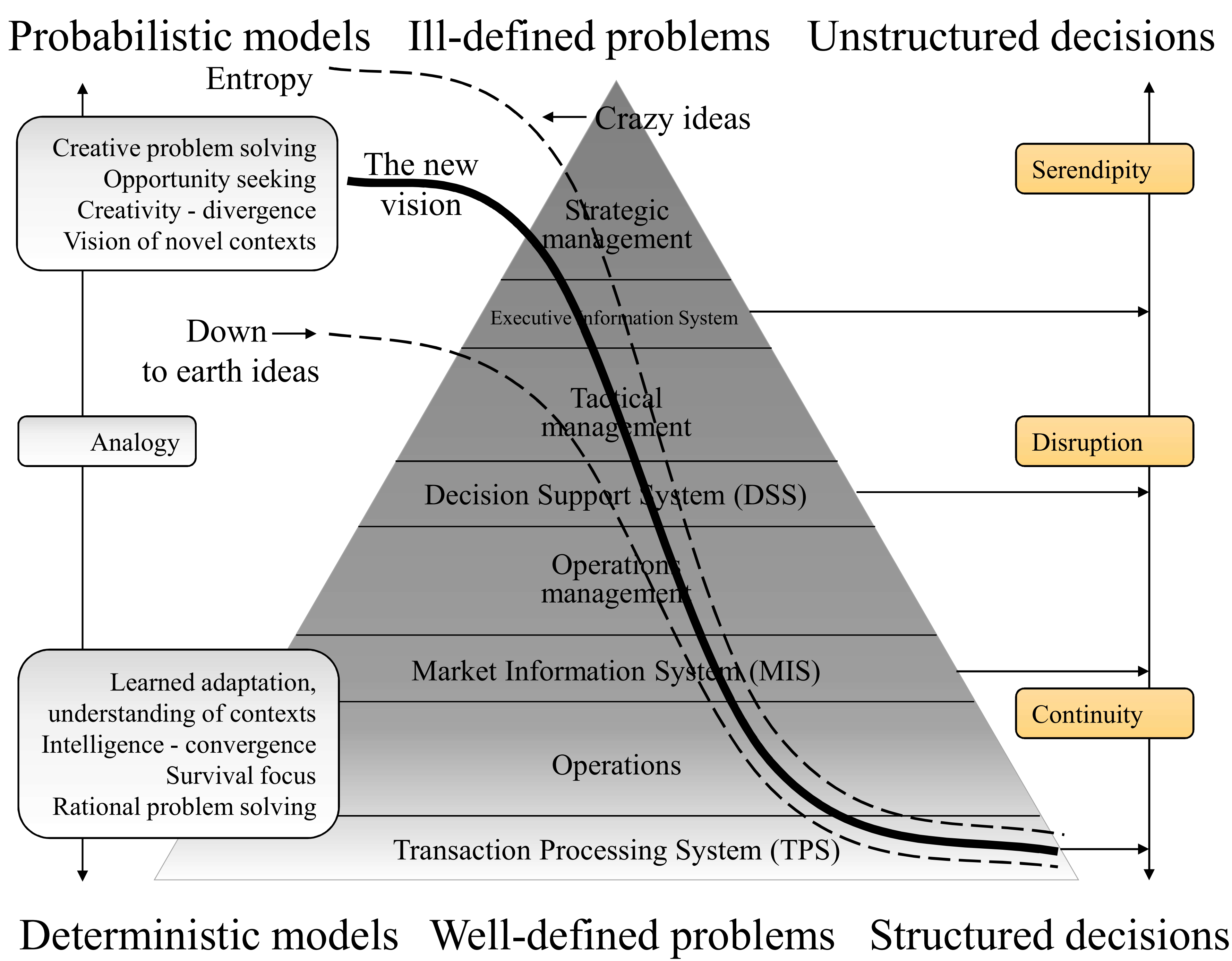
Down
to earth ideas

Mouse

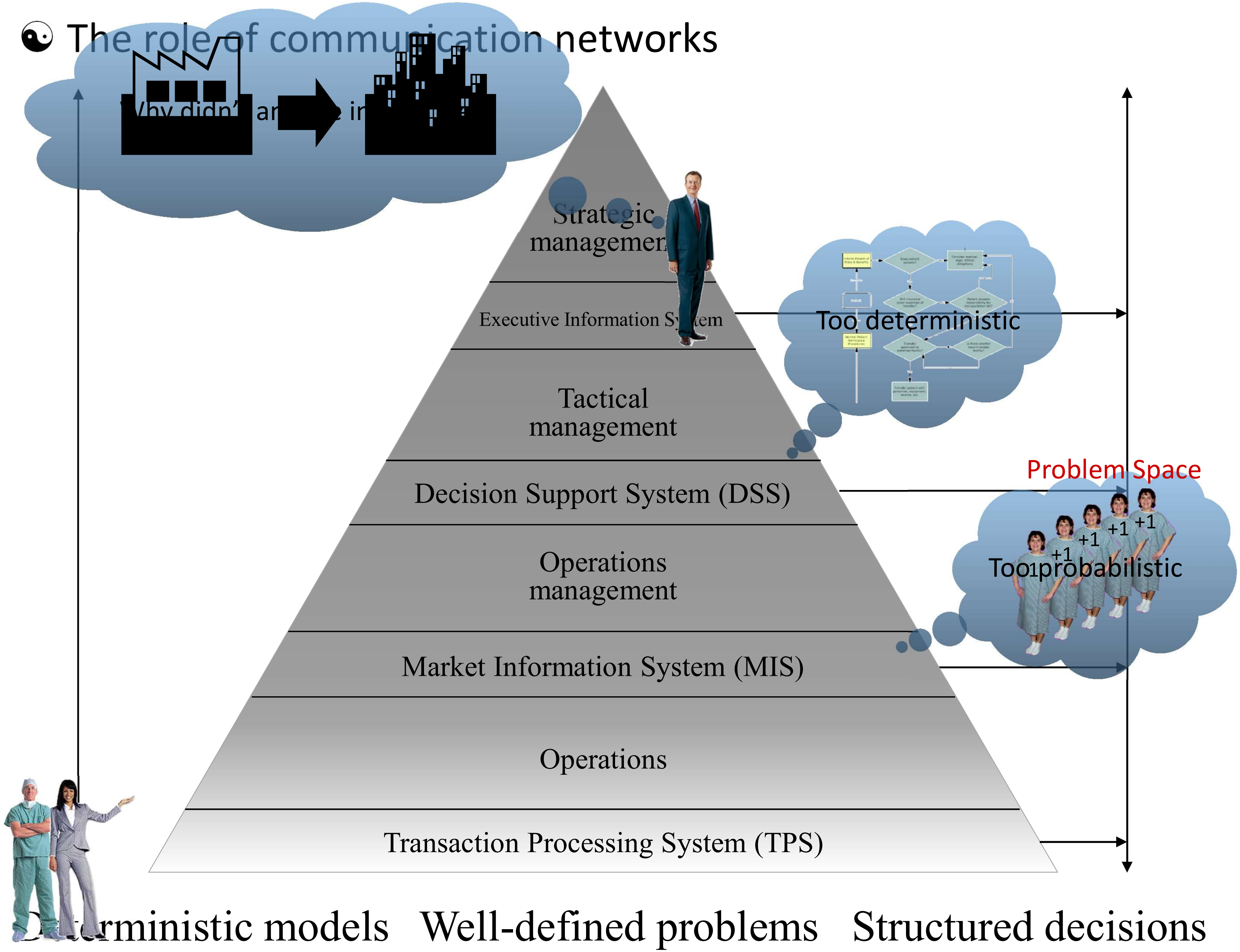
Flip it!

Robotic mouse assembly

Computer mouse assembly using a
compact, highly dexterous 6-axis
robot (Courtesy of ABB Robotics)

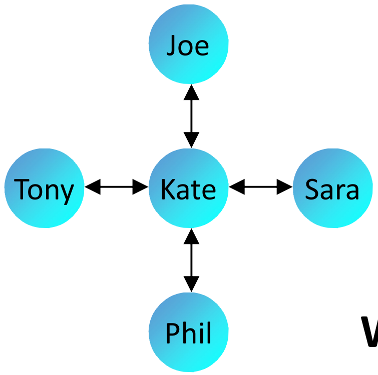

Probabilistic models
Ill-defined problems
Unstructured decisions
Entropy
Crazy ideas
The new vision
Creative problem solving
Opportunity seeking
Creativity - divergence
Vision of novel contexts
Serendipity
Strategic management
Executive Information System
Down to earth ideas
Tactical management
Analogy
Disruption
Decision Support System (DSS)
Operations management
Market Information System (MIS)
Continuity
Learned adaptation, understanding of contexts
Intelligence - convergence
Survival focus
Rational problem solving
Operations
Transaction Processing System (TPS)
Deterministic models
Well-defined problems
Structured decisions

The role of communication networks



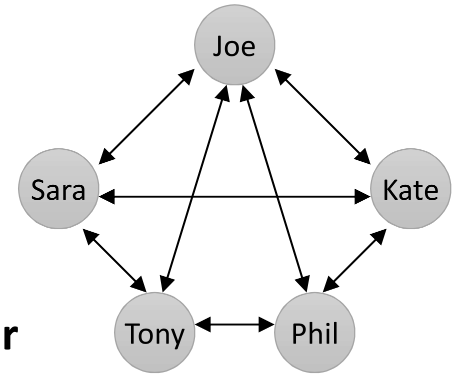
terministic models   Well-defined problems   Structured decisions

☯ The role of communication networks

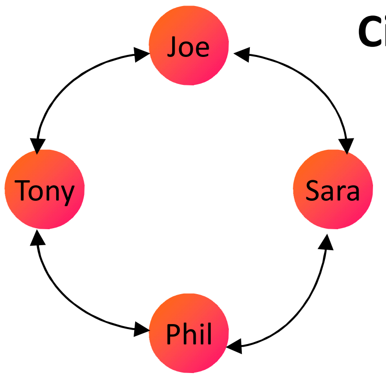


Wheel

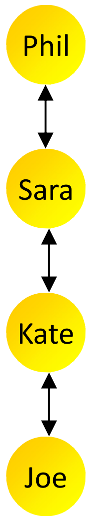
Star



Circle

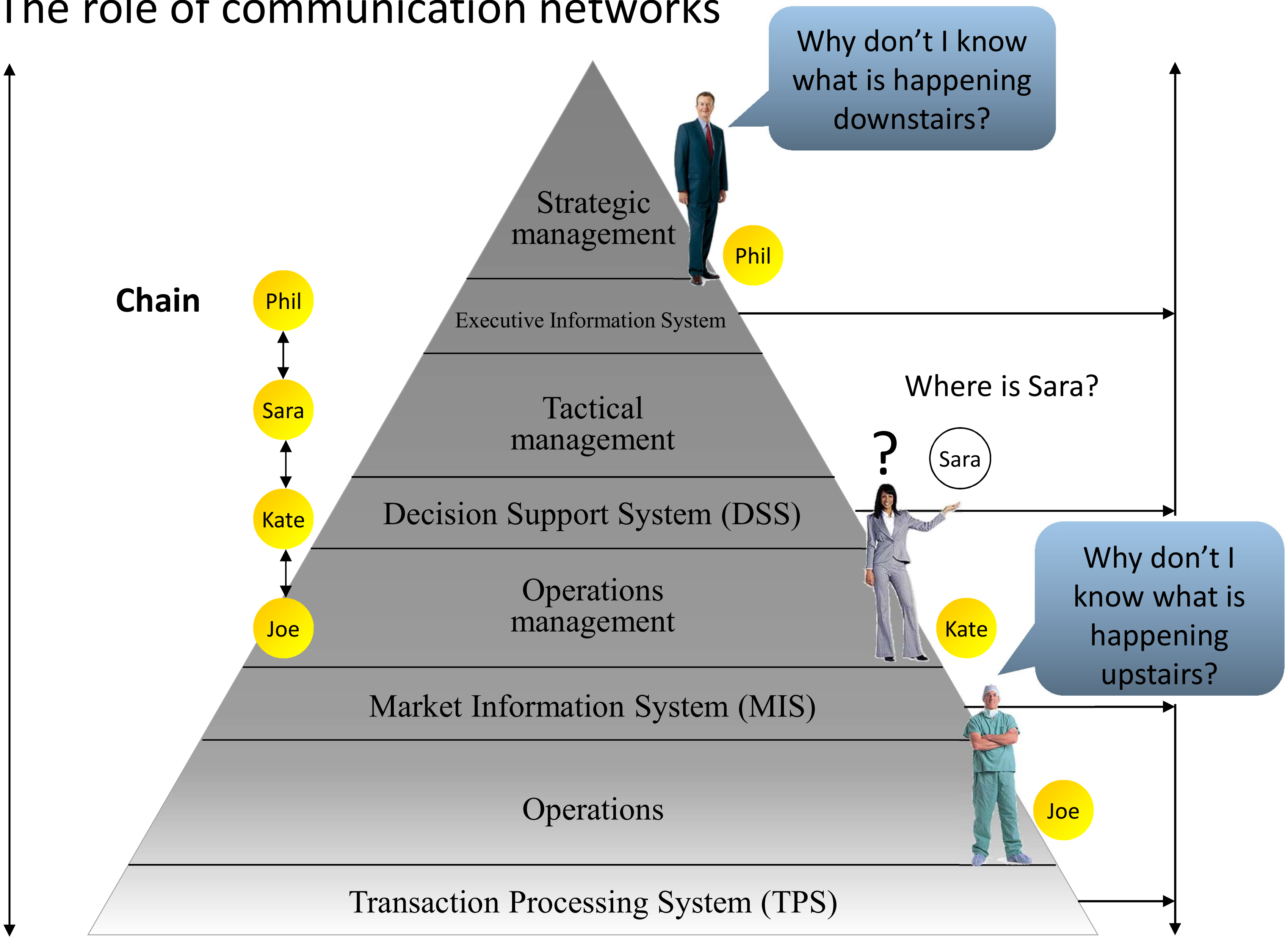


Chain



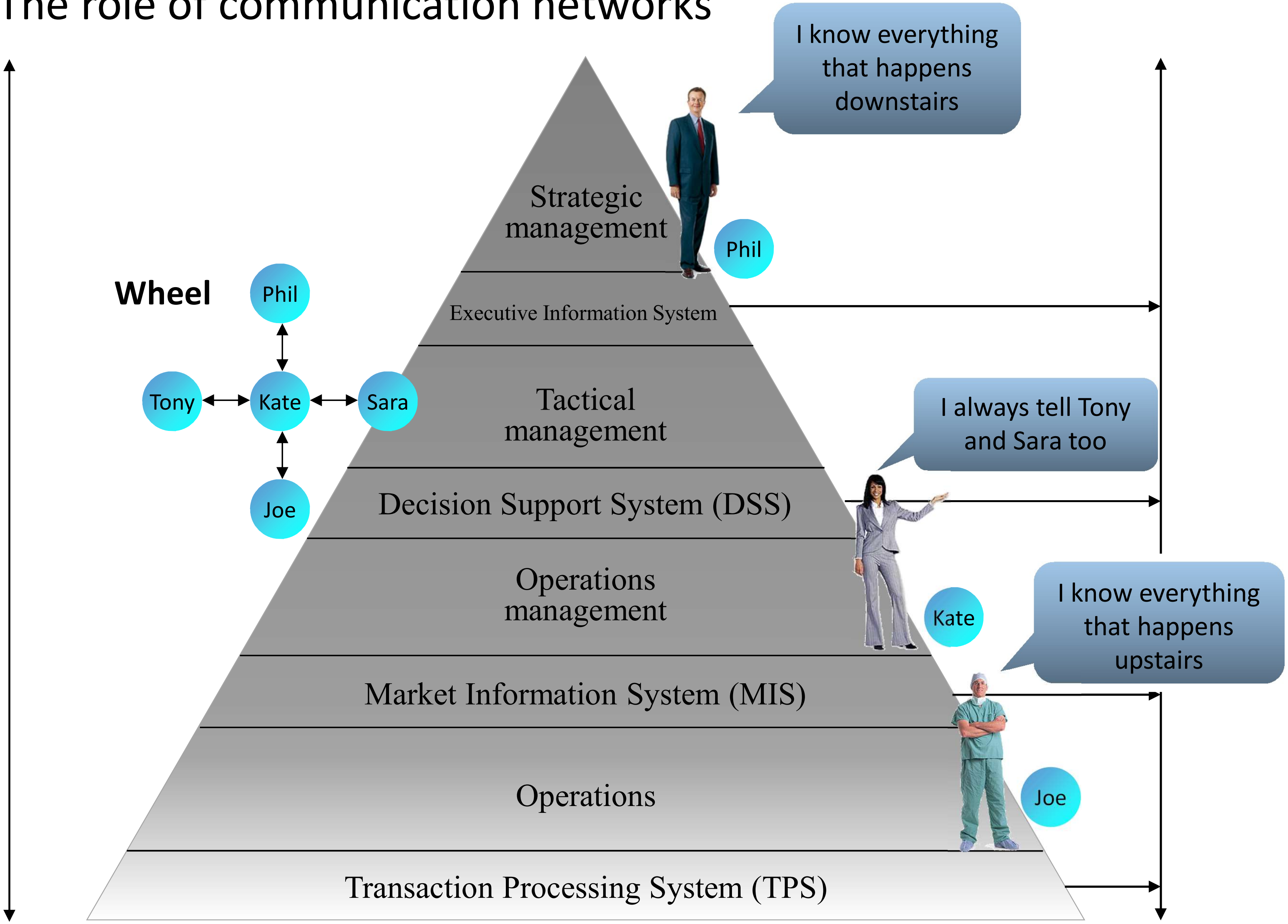
Leavin (1972)

☯ The role of communication networks



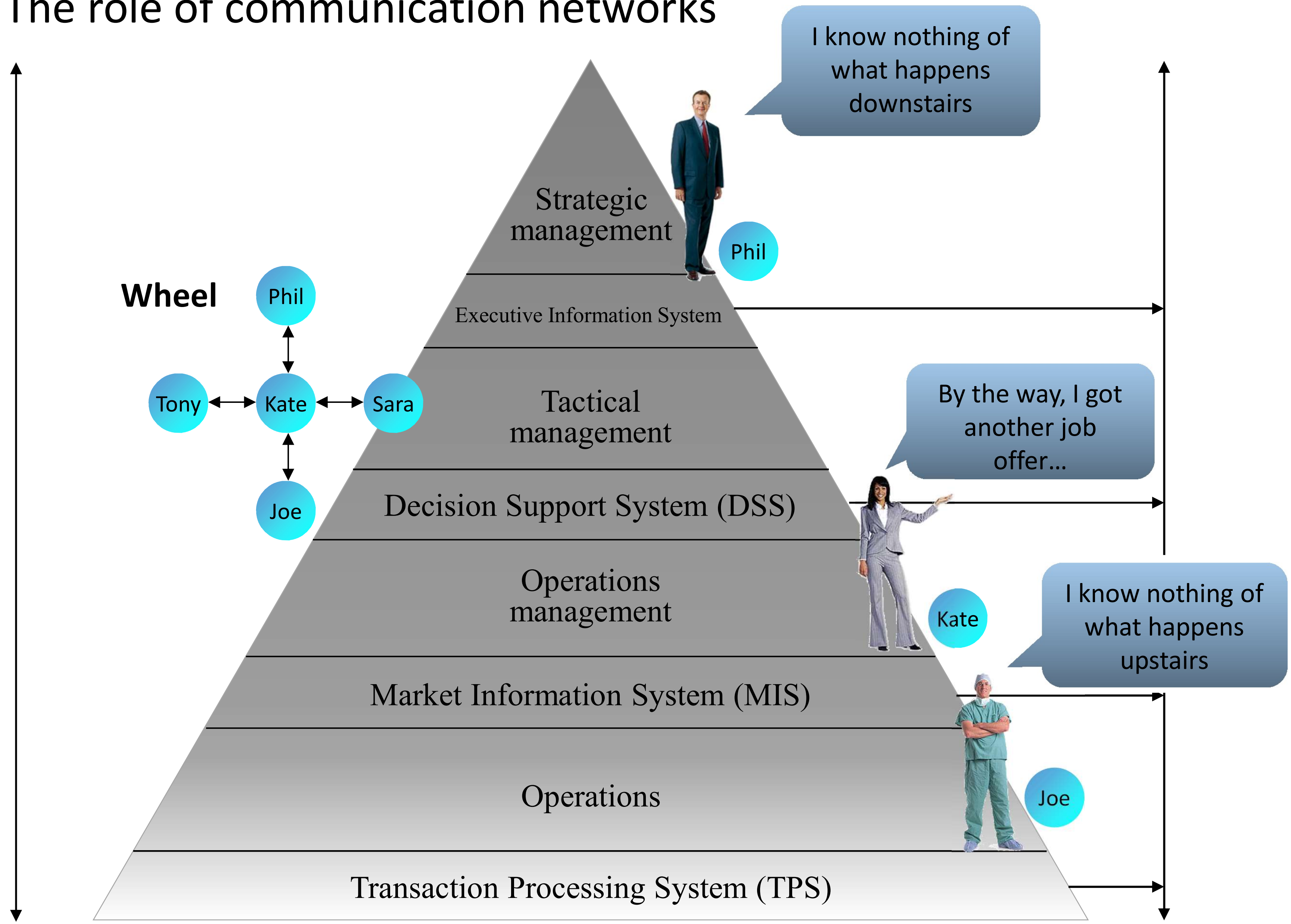
Deterministic models   Well-defined problems   Structured decisions

☯ The role of communication networks



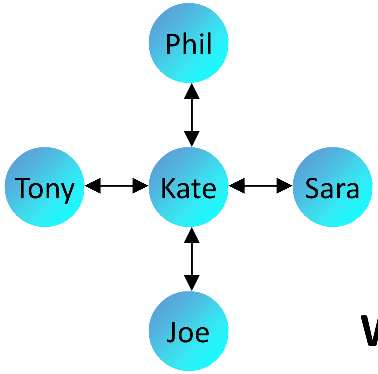
Deterministic models   Well-defined problems   Structured decisions

☯ The role of communication networks



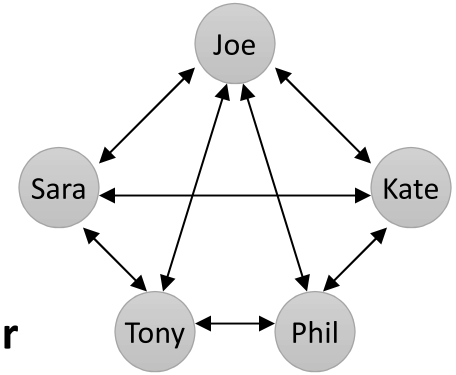
Deterministic models   Well-defined problems   Structured decisions

☯ Analyze a number of real-life cases with me

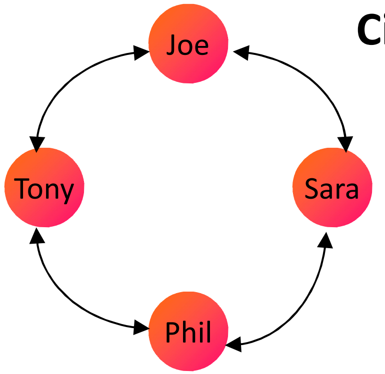


Wheel

Star



Circle



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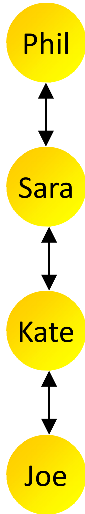
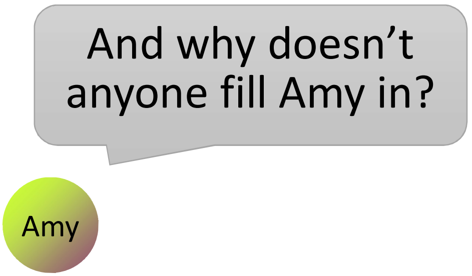
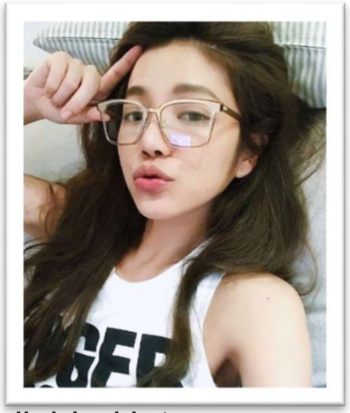




Tencent designer Amy

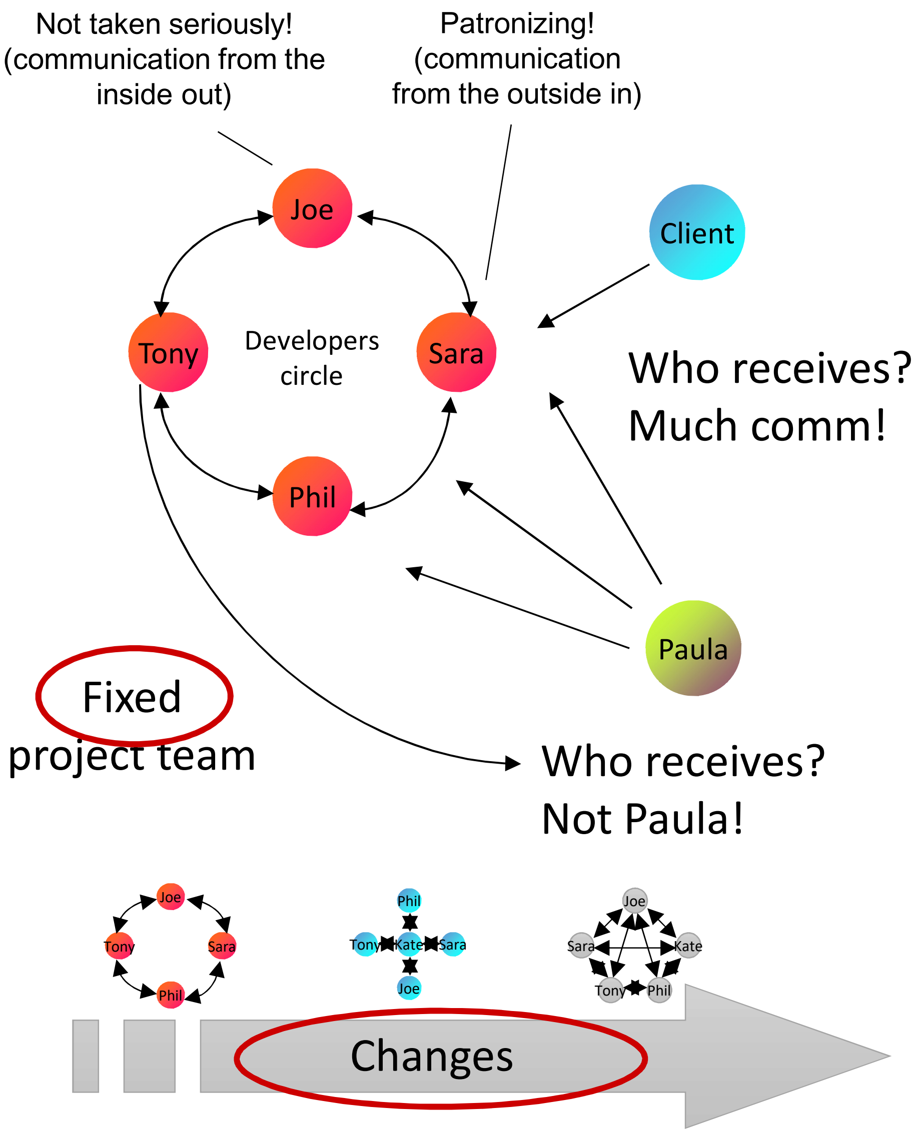


Internet project is divided into three subprojects. I am working within one of those sub-projects. The project team has a project leader for each sub-project. There is a sub-project meeting every week.

Periodically, the three project leaders meet with the client. Problem is communication. Despite the weekly client consultation, things are not done or done differently.

Causes unnecessary confusion. Straightening out of what goes wrong takes a lot of time and energy, and is bad for planning.

☯ Analyze a number of real-life cases wit



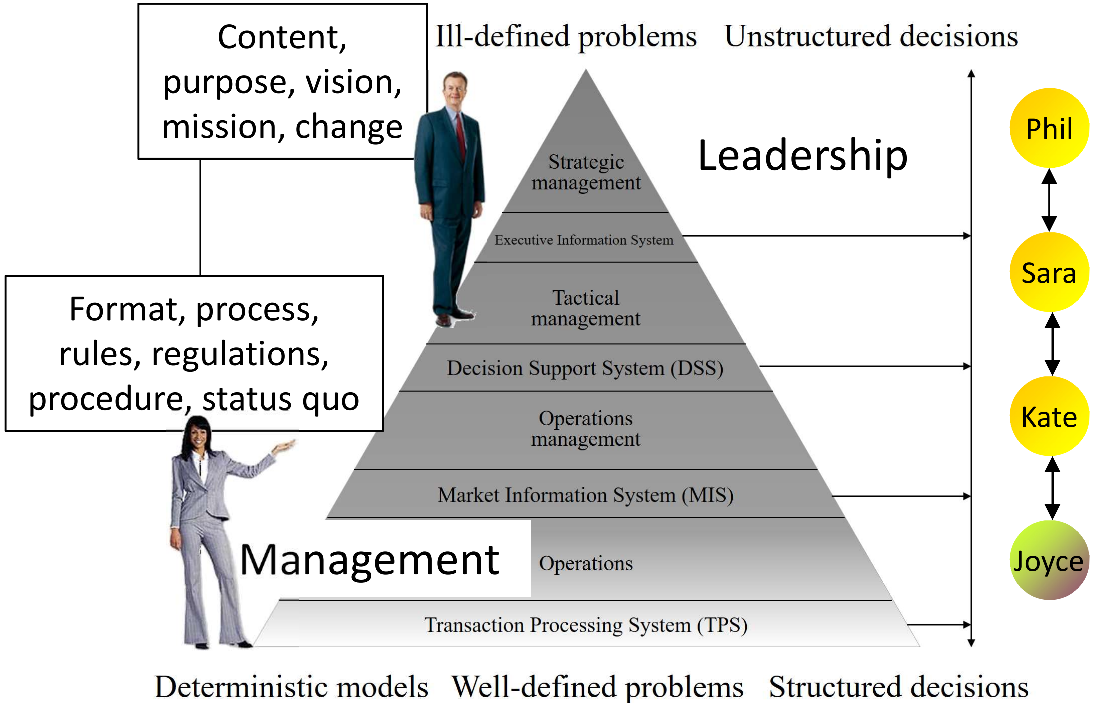
Paula (Chi Nan), Alibaba Fintech engineer

Domain: Adaptations and extensions to a product-service system with a fixed project team with quite different characters. Nature of the problem: The block-chain developers in particular I had to approach ‘cautiously.’

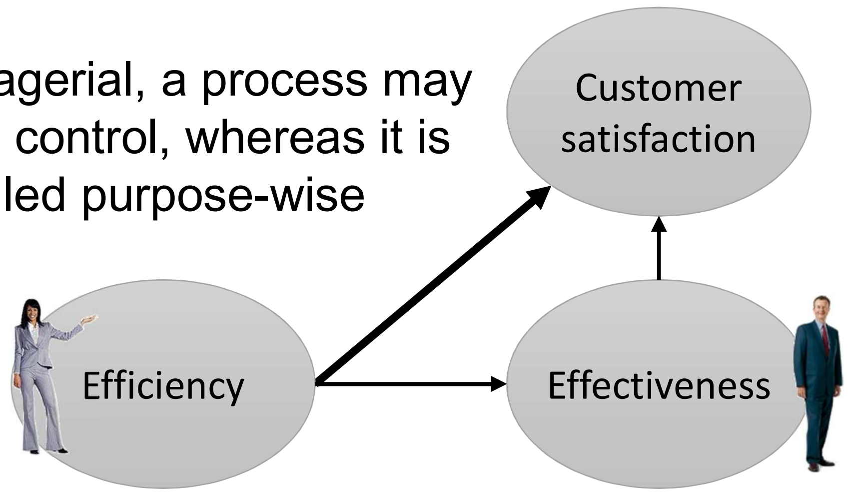
Occasionally, they felt too much patronized on the one hand (I worked designs out in too much detail); on the other hand they felt they were not heard (they came up with ideas that did not always apply and therefore those ideas were not worked out).

There was also difficulty with the flexible and dynamic work environment and customer wishes. This caused tension between the project members. Involved: FO staff, developers, software testers, and PR officers

☯ Analyze a number of real-life cases with me



Managerial, a process may be in control, whereas it is derailed purpose-wise



Joyce (Meifeng), admin systems

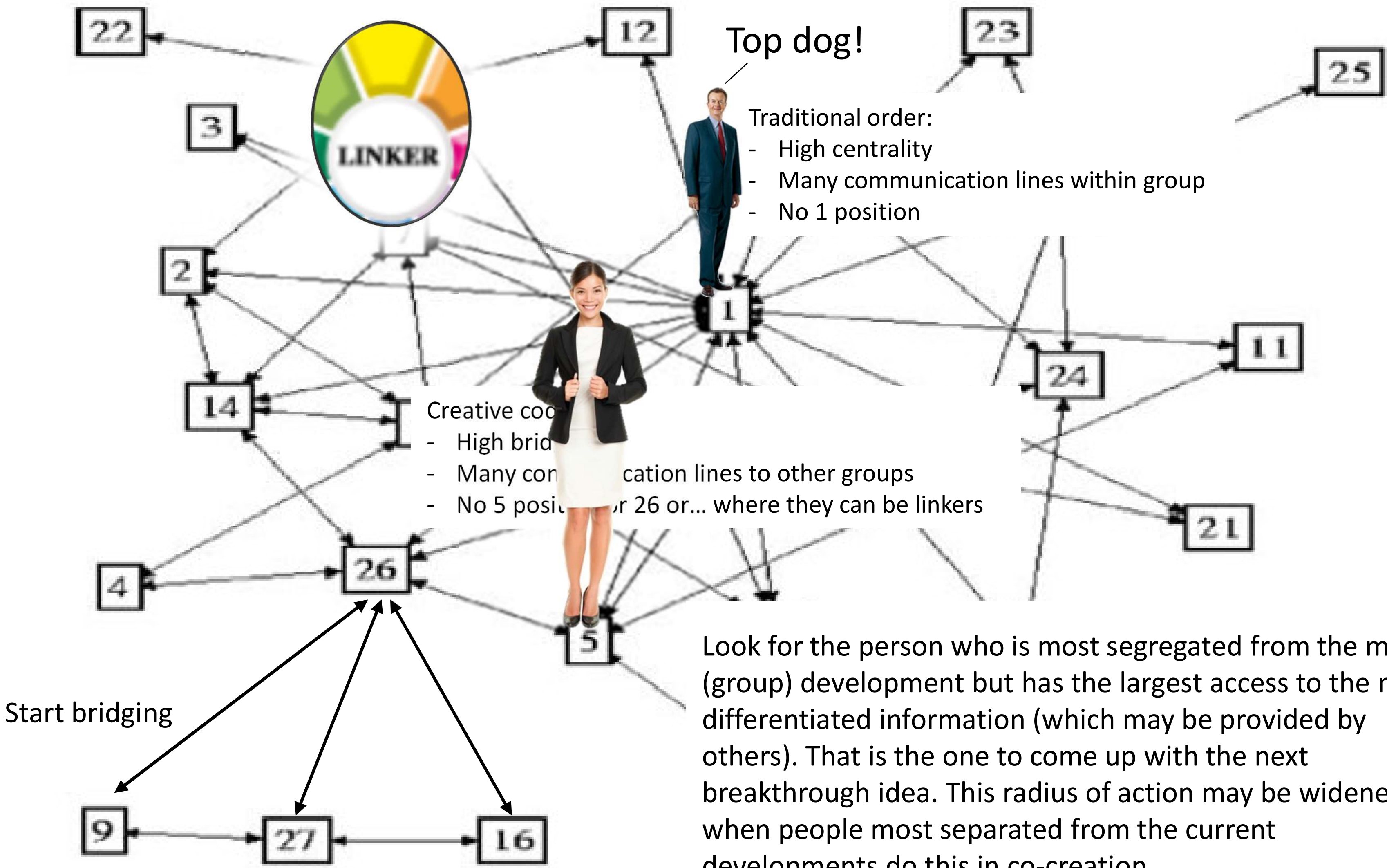
In 2020, I was briefly involved as a manager of e-Government, Basic Registration Addresses and Buildings under the responsibility of the Housing Ministry and Land Registry.

I started planning a new system with analysis. Attempted to draft an Urban Information document. Main bottlenecks:

1. Positioning: Current situation, problem definition, added value of the design unclear
2. Who are stakeholders and what is their role? How to determine functionality and priority?
3. Previous infrastructure decisions to divide maintenance over separate centers and the provision of support.

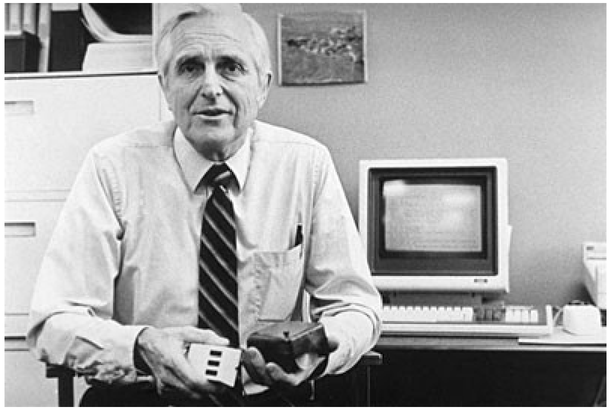
You will encounter this with more, if not all, basic planning projects. The question is whether this was a correct infra-structure and architectural choice?

☯ Where should a creative coder position him/herself?

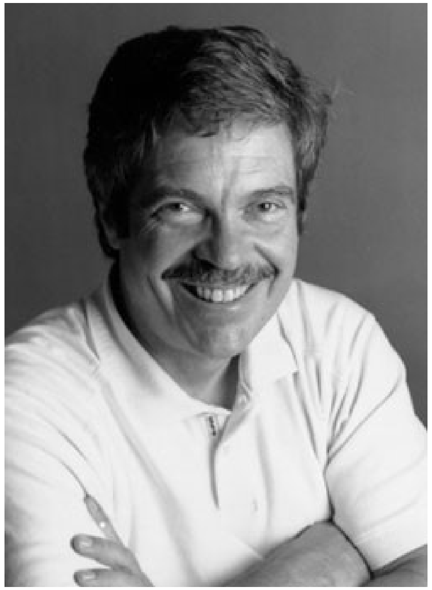


Look for the person who is most segregated from the mean (group) development but has the largest access to the most differentiated information (which may be provided by others). That is the one to come up with the next breakthrough idea. This radius of action may be widened when people most separated from the current developments do this in co-creation

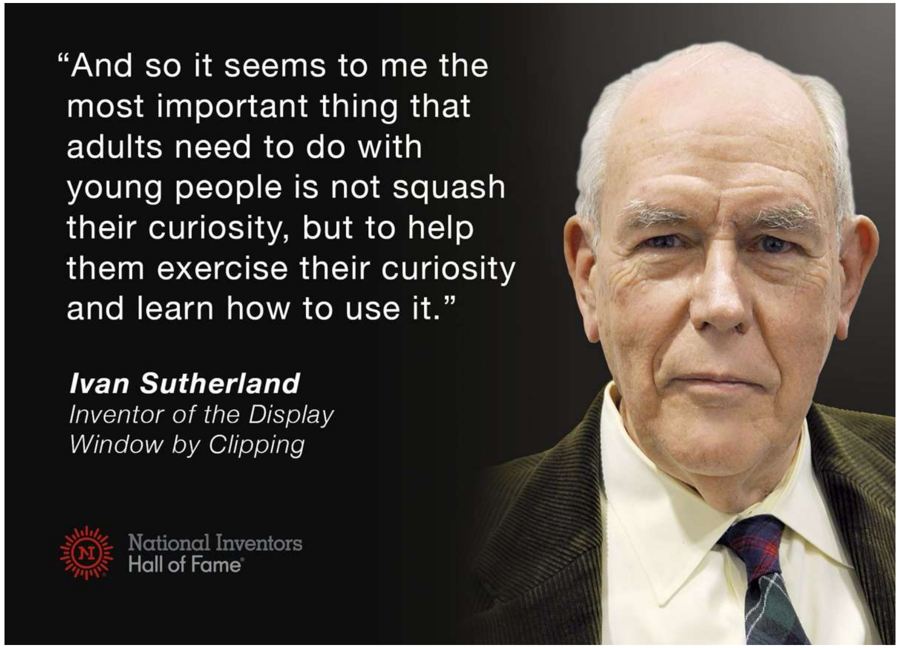
☯ Where should a creative coder position him/herself?



Douglas Engelbart



Alan Kay



Ivan Sutherland

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Creative Coders in Organizations

How to make these people work together?

Creative Coders in Organizations

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Creative Coders in Organizations

How to make these people work together?

Creative Futures

HCI: Q&A and Summary

Johan F. HOORN and Richard LI

Next: Human Information Processing